PassFrame tuition

Statistics & Averages

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 18

2. 27

3. 17

4. 21

5. 17

6. 28

7. 44

8. 70

9. A — 26

10. 112

11. 74 ((68+95+52+81)÷4 = 296÷4)

12. A, B and D (42 pupils were surveyed in total, The mode number of pets owned is 2, More pupils own 2 pets than own 3 pets)

13. A — 74

14. 18

15. 49

Standard

1. 22

2. 48

3. 108

4. A — 26

5. 71

6. 49 ((48+63+31+54)÷4 = 196÷4)

7. 15

8. 88 (100 pets originally, minus 12 = 88)

9. A and B (68 pupils were asked in total, Maths is the mode (most popular))

10. 32.5

11. A — 63

12. 23

13. 21 (football = 42, swimming = 21, other = 84 − 42 − 21 = 21)

14. A — 50

15. 53

Standard / Hard

1. 52

2. 66

3. 50 (football = 40, swimming = 30, other = 120 − 40 − 30 = 50)

4. A — 17

5. 69

6. Mode = 38 (appears twice). Mean = (32+38+26+44+38+50)÷6 = 228÷6 = 38.

7. A, B, C and D (The mean daily sale is £108, The range of sales is £70, Thursday had the highest sales, Wednesday had the lowest sales)

8. 40

9. 54° (15% of 360)

10. A — 68

11. 96

12. 30 (afternoon = 12, mean = (48+12)÷2 = 60÷2)

13. 90

14. 447cm (ordered: 386, 412, 447, 458, 501)

15. A — 26

Homework — Answers

Easy / Standard

1. 20

2. 31

3. 19

4. 23

5. 19

6. 30

7. 48

8. 70

9. A — 28

10. 124

11. 80 ((74+101+58+87)÷4 = 320÷4)

12. A, B and D (47 pupils were surveyed in total, The mode number of pets owned is 2, More pupils own 2 pets than own 3 pets)

13. A — 80

14. 20

15. 49

Standard

1. 26

2. 52

3. 124

4. A — 30

5. 71

6. 53 ((52+67+35+58)÷4 = 212÷4)

7. 19

8. 131 (116 pets originally, plus 15 = 131)

9. A and B (80 pupils were asked in total, Maths is the mode (most popular))

10. 36.5

11. A — 63

12. 27

13. 22 (football = 44, swimming = 22, other = 88 − 44 − 22 = 22)

14. A — 54

15. 53

Standard / Advanced

1. Total = mean × number of days = £84 × 6 = £504.

2. Largest = smallest + range = 15 + 9 = 24.

3. Total = 26 × 4 = 104. Sum of three known numbers = 74. Fourth number = 104 − 74 = 30.

4. Let orange = x, purple = 5x. x + 5x = 360 → 6x = 360 → x = 60°.

5. In an ordered list of 7 numbers, the median IS the 4th (middle) value, so the missing number is 33.

6. Mean = (50+35+45+30+55+25)÷6 = 240÷6 = 40. Range = 55 − 25 = 30.

7. Walk = 40% of 150 = 60. Cycle = 24% of 150 = 36. Car = 150 − 60 − 36 = 54.

8. A — 36 (total was 34×9=306, remove 18 to get 288, 288÷8=36)

9. 80 (Car + Bus = 160° out of 360° = 4/9 of 180 pupils)

10. A, B and C (The total of all 32 scores is 2,272, At least one score is below 71, The highest score minus the lowest score is 48)

11. Chocolate angle = 360 − 160 − 110 − 50 = 40°. Chocolate weight = 40/360 × 1,080g = 120g.

12. Total = 38 × 6 = 228. Sum of five known numbers = 184. Sixth number = 228 − 184 = 44.

13. A — The total of the 10 numbers is 150

14. Class A total = 30 × 1.8 = 54. Class B total = 30 × 1.2 = 36. Combined = 90 pets over 60 pupils = mean of 1.5.

15. Grade A = 120/360 × 60 = 20. Grade C = 60/360 × 60 = 10. Difference = 10.